

LEXINGTON WHALEN

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PROFILE

Machine learning researcher at the intersection of frontier AI and U.S.–Japan technology cooperation, specializing in efficient LLM training and inference. Co-developed NVIDIA’s first diffusion language model series (Nemotron-Labs-Diffusion, Efficient-DLM), then led pretraining and inference optimization for SoftBank SB Intuitions’ Sarashina LLMs. Joining NVIDIA Research (DLER) in November 2026. Publications at EMNLP, ICML, CVPR, and ACL; NSF Graduate Research Fellow.

EDUCATION

Georgia Institute of Technology

M.S. in Computer Science, Specialization in Artificial Intelligence
NSF Graduate Research Fellow.

Atlanta, GA

Expected Spring 2027

University of South Carolina, Honors College

Accelerated B.S./M.S. in Computer Science
Outstanding Senior, Class of 2023.

Columbia, SC

Jan. 2021 – Aug. 2024

PEER-REVIEWED PUBLICATIONS

- [1] Y. Fu, **L. Whalen**, A. Garg, C. Wu, M. Khadkevich, N. Oswald, E. Xie, D. Egert, S. Turuvekere Sreenivas, S. Diao, C. Yu, Y. Yu, W. Chen, S. Norouzi, S. Lan, L. Zhu, J. Wang, J. Jiang, M. Mardani, M. Maghoumi, S. Han, A. Jukić, N. Tajbakhsh, J. Kautz, P. Molchanov. “Nemotron-Labs-Diffusion: A Tri-Mode Language Model Unifying Autoregressive, Diffusion, and Self-Speculation Decoding.” **EMNLP**, 2026. [NVIDIA]
- [2] Y. Fu*, **L. Whalen***, Z. Ye, X. Dong, S. Diao, J. Liu, C. Wu, H. Zhang, E. Xie, S. Han, M. Khadkevich, J. Kautz, Y. (Celine) Lin, P. Molchanov. “Efficient-DLM: From Autoregressive to Diffusion Language Models, and Beyond in Speed.” **ICML**, 2026. [NVIDIA]
- [3] C. Shih, C. Li, C. Yu, H. Fang, S. Li, W. Hsin, **L. Whalen**, H. Suh, G. Eisenhauer, L. Liu, Y. (Celine) Lin. “NeRArch-Sim: A Unified Simulator for Benchmarking and DSE of Neural Rendering Accelerators.” **ISCA**, 2026. [Georgia Tech]
- [4] **L. Whalen***, Z. Du*, H. You*, C. Li, S. Li, Y. (Celine) Lin. “Early-Bird Diffusion: Investigating and Leveraging Timestep-Aware Early-Bird Tickets in Diffusion Models for Efficient Training.” **CVPR**, 2025. [Georgia Tech]
- [5] Z. Ye, Z. Wang, K. Xia, J. Hong, L. Li, **L. Whalen**, C. Wan, Y. Fu, Y. (Celine) Lin, S. Kundu. “LAMB: A Training-Free Method to Enhance the Long-Context Understanding of SSMS via Attention-Guided Token Filtering.” **ACL**, 2025. [Georgia Tech]
- [6] A. Murphy, **L. Whalen**, S. Dubinsky, M. Gavin, J. F. Bailyn, J. Ginn. “On ‘historical unity’ of Russian and Ukrainian: A linguistic perspective on language conflict and change.” **LSA**, Apr. 2023. [University of South Carolina]

* Equal contribution

PREPRINTS & TECHNICAL REPORTS

- [1] **L. Whalen**, Y. Ito, R. Sakamoto. “KBBQ: A Predictive Noise Law and the Limits of Spectrum Flattening in FP4 Quantization.” Technical Report, 2026. [SB Intuitions]
- [2] **L. Whalen**, Y. Ito, R. Sakamoto. “Teaching Diffusion to Speculate Left-to-Right.” Technical Report, 2026. Under review at TMLR. [SB Intuitions]

PATENTS

- [1] **L. Whalen**, Y. Ito, R. Sakamoto. 「拡散ベースの投機的ドラフトモデルにおける受理長改善アルゴリズム」 Japanese Patent (filed), 2026. [SB Intuitions]
“Algorithm to Improve Acceptance Length of Diffusion-Based Speculative Draft Models.”
- [2] Y. Fu, **L. Whalen**, X. Dong, S. Diao, C. Wu, E. Xie, S. Han, J. Kautz, Y. (Celine) Lin, P. Molchanov. “Training Framework for Converting Autoregressive Language Models into Faster Diffusion Language Models.” U.S. Patent (filed), 2026. [NVIDIA]

RESEARCH EXPERIENCE

NVIDIA Research – DLER Team

Research Scientist

Santa Clara, CA

Nov. 2026 –

- Making intelligence efficient.

SoftBank – SB Intuitions

Pretraining & Inference Optimization Research Lead

Tokyo, Japan

Apr. 2026 – Nov. 2026

- Co-lead pretraining and post-training optimization research for the [Sarashina series](#), Japan’s largest LLMs, at SoftBank’s SB Intuitions R&D Division.
- Lead novel [quantization efforts](#), spanning models up to hundreds of billions of parameters.
- Built SB Intuitions’ first [speculative-decoding](#) inference stack for the Sarashina family; introduced training-time drafter-alignment methods that raise the accepted draft length by 21–76% over a position-uniform baseline.
- Serve as a key technical voice in Japanese-language cross-team discussions on pretraining, post-training, and inference optimization, bridging research and engineering, and collaborate with leading Japanese and international technology companies and government agencies to evaluate GPU resources and co-develop long-term compute strategies for large-scale training.

NVIDIA – DLER Team

Efficient Deep Learning Intern & Data Filtering Challenge Lead

Atlanta, GA & Santa Clara, CA

Dec. 2024 – Mar. 2026

Mentors: Dr. [Pavlo Molchanov](#) & [Yonggan Fu](#) (NVIDIA Research).

- Co-developed [Nemotron-Labs-Diffusion](#) (EMNLP 2026), a tri-mode (autoregressive / diffusion / self-speculation) language model family at 3B–14B scale; the 8B model decodes 6× more tokens per forward pass than Qwen3-8B at comparable accuracy (up to 4× higher throughput); [released on Hugging Face](#) with ~1M total downloads.
- Co-first author of [Efficient-DLM](#) (ICML 2026), converting pretrained autoregressive models into diffusion LMs; the 8B model reaches +5.4%/+2.7% higher accuracy at 4.5×/2.7× higher throughput than Dream 7B / Qwen3 4B ([open-weights on Hugging Face](#)); presented directly to CEO Jensen Huang.
- Helped build the training, evaluation, and inference infrastructure for NVIDIA’s first series of diffusion language models, adopted by 5+ internal research teams for vision-language models, embeddings, and latent reasoning.
- Organized NVIDIA’s [2025 International Data Filtering Challenge for Training Edge Language Models](#) across 30+ university and industry teams.

Georgia Institute of Technology

Graduate Researcher

Atlanta, GA

Aug. 2024 – Dec. 2025

- Led a team of 3 Ph.D. and Masters students to cut diffusion-model training time by 2.9–5.8× (up to 10.3× vs. standard train-prune-finetune) while maintaining generation quality; [CVPR 2025](#).
- Algorithm lead for the United States’ [ARPA-H THOR healthcare initiative](#): engineered resource-efficient ML architectures powered by energy from the human body, collaborating with teams from Stanford, Northwestern, Carnegie Mellon, and MIT, reducing cancer detection sensor degradation by over 70% and extending sensor lifetime from a few hours to several days.

HONORS & AWARDS

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| • SoftBank Group Representative, 2026 Entrance Ceremony | Mar. 2026 |
| • NSF Graduate Research Fellowship (\$150,000 value) | Apr. 2024 |
| • Toyo University Exchange Student Representative | Jul. 2023 |
| • University of South Carolina Outstanding Senior , Class of 2023 | Feb. 2023 |
| • U.S. Department of State Critical Language Scholar of Japanese | Jan. 2023 |
| • University of South Carolina Top Scholar (\$100,000+ value) | Aug. 2020 |

INVITED TALKS

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| • “速く、小さく、軽く：LLM効率化のベストプラクティス”, SoftBank – SB Intuitions
<i>Faster, Smaller, Lighter: Best Practices for LLM Efficiency</i> | Oct. 2026 |
| • “拡散言語モデル：基本から応用まで”, SoftBank – SB Intuitions
<i>Diffusion Language Models: From Fundamentals to Applications</i> | Jul. 2026 |
| • 入社スピーチ , SoftBank
<i>Entrance Ceremony Speech</i> | Apr. 2026 |

SERVICE & OUTREACH

- Serve as an informal advisor on U.S.–Japan AI and technology policy, engaging with U.S. Embassy Tokyo and Japanese government stakeholders.

LANGUAGE SKILLS

English (Native) · **Japanese** (Native Level, JLPT N1 Certified) · **Mandarin** (Intermediate)